

AI Travel Recommendation Project — How to Start & Can You Do It Without GPU?

YES — you can develop the entire project without a GPU. Below is the complete roadmap.

PHASE 1 — SETUP

- Frontend: React
- Backend: Flask/Django
- Database: MySQL
- ML/NLP: spaCy + scikit-learn
- No GPU required.

PHASE 2 — BACKEND APIS

- 1) /parse-query → Extracts city, days, budget, travel type.
- 2) /recommend-destinations → Content-based filtering using cosine similarity.
- 3) /generate-itinerary → Rule-based scheduling.

PHASE 3 — FRONTEND

- Home Page: search bar, featured places, weather
- Itinerary Page: show plan, allow editing
- User Profile Page: travel preferences

PHASE 4 — NLP MODULE

Options:

- spaCy small model (recommended)
- DistilBERT (CPU-friendly)
- Rule-based extraction

PHASE 5 — RECOMMENDATION SYSTEM

- Create features: cost, activity type, season, weather

- Compute similarity between user prefs and destinations
- Works fast on CPU

PHASE 6 — INTEGRATIONS

- Weather API (OpenWeatherMap)
- Google Maps API
- Trip Advisor data or own dataset

NO NEED FOR:

- GPU
- Training huge models
- TensorFlow heavy workloads

COMPLETION TIME:

15–20 days for full project.

You can request next:

- Full backend code
- React components
- Database schema
- FYP report
- Proposal, SRS, ERD, UML diagrams